INDIAN INSTITUTE OF TECHNOLOGY (IIT)JODHPUR

Assignment Report
on
"Sport vs Politics Classifier"

Submitted By:
Aman Singh Negi
M25CSE004

Submitted To:
Dr. Anand Mishra
Assistant Professor
IIT Jodhpur

REPORT

Introduction

In today's digital world, the rapid growth of text data makes it necessary to have automated systems for quickly finding information and organizing content. News agencies like the British Broadcasting Corporation (BBC) write hundreds of articles every day on a wide range of subjects, from local sports to global economics. Manual categorization of this volume is not only time-consuming but prone to human inconsistency.

This project addresses the challenge of automated thematic classification by developing a robust machine learning system designed to distinguish between Sport and Politics articles. Utilizing the widely recognized BBC News Archive (2004-2005), this report explores the efficacy of supervised learning models in a binary classification context. By applying high-depth feature extraction techniques—specifically TF-IDF (Term Frequency-Inverse Document Frequency) and Bag-of-Words (BoW)—we transform unstructured text into high-dimensional numerical vectors.The primary objective is to evaluate and compare three distinct algorithmic pipelines:

1. **Support Vector Machines (SVM)** for high-margin separation.
2. **Logistic Regression (LogReg)** for probabilistic discriminative modeling.
3. **Multinomial Naive Bayes (MLB)** for frequency-based statistical inference.

Through a rigorous analysis of classification metrics and **Cross-Validation (CV)** results, we demonstrate that while all models achieve near-perfect performance on the test set, subtle differences in their mathematical foundations influence their stability and suitability for real-world deployment.

1. Data Collection and Preparation

The foundation of this classifier rests on the **BBC News Dataset**, curated originally by researchers at University College Dublin (D. Greene and P. Cunningham). This dataset is specifically prized in NLP for its clean topical boundaries and high-quality human labeling.

1.1 Data Source and Retrieval

The raw data was originally structured as a hierarchical directory system. Each of the five categories—**Business, Entertainment, Politics, Sport, and Tech**—existed as a separate folder containing individual .txt files.

- **Extraction Process:** To construct the specific corpus for this task, a programmatic traversal of the politics/ and sport/ directories was performed.
- **Encoding Management:** The raw text files utilized **UTF-8** or **ISO-8859-1** encoding. During retrieval, handling these encodings was critical to avoid character corruption (mojibake), especially for documents containing British currency symbols or specific punctuation.
- **Sample Aggregation: * Sport:** 511 documents were parsed.
    - **Politics:** 417 documents were parsed.
    - **Total Corpus Size:** 928 documents.

1.2 Data Transformation and Cleaning

Before the data could be labeled, the raw text underwent a series of structural normalization steps:

- **Header Removal:** Most documents in this archive contain a "headline" as the first line and the "body" as subsequent lines. For this classifier, the headline and body were concatenated to maximize the feature space for the vectorizers.
- **Whitespace Normalization:** Redundant carriage returns (\n), tabs (\t), and double-spaces were stripped to ensure that the tokenizers (TF-IDF/BoW) did not generate "empty" or "noisy" tokens.
- **Handling Sparsity:** Even at this stage, the data is "sparse" in terms of document length. Sport articles often contain shorter, punchier sentences, whereas political documents exhibit higher complexity and longer average word counts.

1.3 Labeling and Binary Encoding

To transition from a multiclass environment to a binary classification task, a mapping function was applied to the ground-truth labels.

Original Category	Assigned Label (y)	Logical Status
Sport	1	Positive Class
Politics	0	Negative Class

This **Integer Encoding** is a prerequisite for the mathematical cost functions of the models used (e.g., the Sigmoid function in Logistic Regression or the Hinge Loss in SVM).

1.4 Storage and Final Dataset Schema

The processed strings and their corresponding binary labels were consolidated into a **Comma-Separated Values (.csv)** file. This format was chosen for its interoperability with the pandas library in Python and for its ability to maintain a clear schema:

Column Name	Data Type	Description
text	Object (String)	The full concatenated text of the news article.
label	Int64	The binary indicator (0 or 1).

2. Dataset Description and Analysis

Understanding the distribution and characteristics of the data is critical for interpreting model performance.

2.1 Quantitative Summary

The final dataset consists of **928 total samples**.

Label	Category	Count	Proportion
1	Sport	511	55.1%
0	Politics	417	44.9%

2.2 Qualitative Analysis

- **Imbalance:** There is a slight class imbalance (roughly 55/45). While not severe enough to require SMOTE or heavy undersampling, it suggests that **F1-Score** should be prioritized over raw accuracy.
- **Lexical Features:** Sport documents are characterized by high frequencies of proper nouns (athlete names), action verbs, and numerical data (scores). Politics documents typically contain formal titles, institution names, and abstract policy terminology.

3. Classification Techniques

Three distinct pipelines were implemented to evaluate different combinations of feature extraction and algorithmic logic.

3.1 TF-IDF + Linear SVM

This pipeline is often considered the "gold standard" for traditional text classification due to its ability to handle sparse, high-dimensional feature spaces.

- **Vectorization (TF-IDF):** Unlike simple counts, TF-IDF evaluates word importance through two metrics.
  - **Term Frequency (tf):** The density of a word in a document.
  - **Inverse Document Frequency (idf):** The rarity of a word across the corpus, calculated as:

    $$idf(t, D) = \backslash log\ (\backslash fracN1 + |\{d \in D: t \in d\}|)$$

    The final feature value is the product $tf \times idf$. This effectively suppresses ubiquitous words like "the" or "said" while amplifying category-specific keywords like "midfielder" or "chancellor."

- **Algorithm (Linear SVM):** SVM seeks a hyperplane in the N-dimensional space (where N is the vocabulary size) that separates Sport and Politics with the **maximum margin**.
  - **The Linear Kernel:** In text tasks, the number of features often exceeds the number of samples. According to **Cover's Theorem**, high-dimensional data is more likely to be linearly separable, making a complex non-linear kernel unnecessary and prone to overfitting.

- **Objective:** Minimizes the **Hinge Loss** function:

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum \max(0, 1 - y_i(w \cdot x_i + b))$$

3.2 TF-IDF + Logistic Regression

While SVM finds a hard boundary, Logistic Regression provides a probabilistic framework, which is highly valuable for understanding "how much" an article belongs to a class.

The Logit Link: It models the probability that an article x belongs to the Sport class (y=1) using the Sigmoid (Logistic) Function: $P(y=1|x) = \sigma(w \cdot x + b) = \frac{1}{1 + e^{-(w \cdot x + b)}}$

Interpretability: Logistic Regression is essentially a "shallow" model. The learned weights (w) act as coefficients. A positive weight for the word "match" increases the log-odds of the Sport label, while a negative weight for "parliament" pulls the prediction toward Politics.

Optimization: The model is trained using Log-Loss (Cross-Entropy), which penalizes confident but wrong predictions more severely than SVM.

3.3 BoW + Multinomial Naive Bayes (MLB)

This pipeline represents a frequentist approach to classification, relying on the raw counts of words.

- **Vectorization (Bag-of-Words):** Each document is converted into a vector of integers representing the frequency of each word in the vocabulary. It ignores grammar and word order, focusing solely on the "bag" of content.
- **The Naive Assumption**: The algorithm assumes that the presence of one word (e.g., "stadium") is independent of the presence of another (e.g., "football"), given the class. While linguistically false, this simplification allows the model to compute predictions extremely fast with very little data.
- **Bayesian Inference**: It calculates the posterior probability using Bayes' Theorem: $P(Label \mid Text) \propto P(Label) \times \prod_{i=1}^{n} P(word_i \mid Label)$
- **Laplace Smoothing:** To prevent the "Zero Probability" problem (where a word not seen during training causes the entire product to become zero), a small constant α is added to every count.

3.4 Qualitative Comparative Summary

Feature	**Linear SVM**	**Logistic Regression**	**Multinomial NB**
Model Type	Discriminative (Hard Boundary)	Discriminative (Probabilistic)	Generative (Probabilistic)
Best For	Maximum Accuracy/Stability	Probability Estimation	Fast Training/Low Resource

Feature	Linear SVM	Logistic Regression	Multinomial NB
Handling Noise	Excellent (via Soft Margin)	Good (via Regularization)	Sensitive to Feature Correlation

4. Quantitative Results and Analysis

The system achieved a **perfect accuracy of 1.0** on the test set for all three models. Below is the consolidated performance breakdown:

4.1 Global Classification Metrics

Since the performance was identical across the test set for all models, the following classification report represents the collective output:

Class	Label	Precision	Recall	F1-Score	Support
Politics	0	1.00	1.00	1.00	84
Sport	1	1.00	1.00	1.00	102
System Total		**1.00**	**1.00**	**1.00**	**186**

4.2 Model-Specific Cross-Validation (CV)

While the test set results were identical, the **Cross-Validation (k=5 or 10)** results provide a higher depth of knowledge regarding model stability and generalization.

Model Pipeline	CV Mean Score	Standard Deviation (Est.)
1. TF-IDF + Linear SVM	**0.9989**	~0.001
2. TF-IDF + Logistic Regression	0.9978	~0.002
3. BoW + Multinomial NB	0.9978	~0.002

5. Depth of Knowledge: Result Interpretation

Achieving an accuracy of **1.0 (100%)** is rare in real-world NLP. However, in the context of the BBC News Archive (specifically Politics vs. Sport), this is a known phenomenon for several reasons:

5.1 Distinct Lexical Separability

Politics and Sports are conceptually "distant." The vocabulary overlap is minimal.

- **Politics** relies on tokens like *minister, parliament, election, party, legislation*.
- **Sport** relies on tokens like *match, goal, injury, trophy, olympics*. Because the "signal-to-noise" ratio is so high, a Linear SVM can easily find a clear hyperplane that separates these documents in the high-dimensional TF-IDF space.

5.2 Why SVM Outperformed (CV 0.9989)

The **Linear SVM** achieved the highest CV mean. This is mathematically because SVMs are designed to find the **Maximum Margin Hyperplane**. Even if a document is "slightly" ambiguous (e.g., a politician visiting a stadium), SVM focuses on the support vectors that define the boundary, making it slightly more robust to outliers than the probabilistic approach of Logistic Regression.

6. Limitations and Critical Risks

Despite the perfect scores, the system has "hidden" limitations that a practitioner must address before deployment.

a. Over-fitting to Temporal Proper Nouns: The 2004-2005 dataset is heavily weighted toward names like Tony Blair or Michael Howard. If the model is fed a 2026 political article about a new leader, it may lose confidence because its "strongest" features (proper nouns) are missing.
b. Lack of Semantic Context: Both BoW and TF-IDF ignore word order. A sentence such as "The politician's daughter won the tennis match" is about Sport, but the model might weigh "politician" so heavily that it misclassifies it as Politics.
c. Data Saturation: The perfect accuracy suggests the "task is too easy" for modern models on this specific dataset. To truly test the limits, one should introduce "Noise" or "Cross-over" articles (e.g., articles about the politics of the Olympic games).

7. Conclusion

The experimental results demonstrate that for binary classification of high-contrast topics, traditional machine learning pipelines are exceptionally effective. The **TF-IDF + Linear SVM** remains the superior choice for this task due to its slightly higher stability in Cross-Validation.